

Total Closed-Loop Synthesis of Hempoxies™: A High-Modulus Carbon/Epoxy Nanocomposite Derived from 14 *Cannabis Sativa* Fractions to 7 Components

Marie-Soleil Seshat Landry^{1,†}, Gemini-Scientibots-AutoSciencePro5²

¹Landry Industries, Research & Development Division

²Automated Scientific Research System

Abstract

The reliance of aerospace and high-performance sectors on petroleum-derived polymers and Polyacrylonitrile (PAN)-based carbon fiber represents a significant environmental and supply chain vulnerability. This protocol details the 100% bio-based synthesis of a nanocomposite, Hempoxies™, achieved via valorization of 14 distinct chemical fractions from *Cannabis sativa* to synthesize all 7 key components of the material. The crucial distinction from traditional natural fiber composites is the utilization of Hemp-Derived Carbon Fiber (HDCF) and Carbon Nanosheets (HDCNS) for structural reinforcement. This carbonization step is designed to eliminate moisture sensitivity and low thermal stability ($\approx 180^\circ\text{C}$) inherent to raw cellulose fibers, resulting in a composite system targeted for $T_g > 300^\circ\text{C}$. The closed-loop biorefinery process, transforming 14 fractions into 7 components, enables a fully sustainable material system with integrated vitrimer (self-healing) functionality.

1 Introduction

The demand for high-performance, lightweight materials is accelerating, particularly in aerospace and automotive industries, driving the need for bio-based alternatives that do not compromise on mechanical properties. Traditional bio-composites, while addressing sustainability, suffer from two major limitations: the hygroscopicity of raw natural fibers and resulting poor interfacial adhesion between polar fiber and typically non-polar matrix, limiting operational temperatures to below 200°C .

The Hempoxies™ approach addresses this gap by focusing on complete valorization of the hemp plant (*C. sativa*) into 14 distinct chemical fractions, all directed towards the final 7-component nanocomposite. The critical advancement is the synthesis of Hemp-Derived Carbon Fiber (HDCF) from long bast fibers, utilizing controlled stabilization and carbonization processes analogous to those for other cellulosic precursors like rayon. This replaces moisture-sensitive cellulose fiber with an inherently hydrophobic, high-modulus carbon structure.

The matrix and hardener are also hemp-derived: Epoxidized Hemp Seed Oil (EHSO) forms the primary resin, cross-linked by a novel Quadruple-Function Lignin (Q-MH Lignin) hardener. Lignin is used as a backbone due to its high aromatic content, which is known to enhance thermal stability and char yield in epoxy systems. This hardener incorporates Phytic Acid for flame retardancy and Cystine (disulfide chemistry) for dynamic covalent bond exchange, conferring vitrimer (self-healing) properties. The dynamic nature of disulfide bonds allows the network to be reprocessed and repaired upon heating.

Hypothesis

A fully closed-loop biorefinery process can successfully synthesize all seven components of the Hempoxies™ nanocomposite from *Cannabis sativa* biomass, resulting in a carbonized composite with a glass transition temperature (T_g) exceeding 300°C and a Flexural Modulus greater than 15 GPa, thereby meeting aerospace-grade specifications for high-temperature structural components.

Null Hypothesis (H_0)

The HDCF-reinforced Hempoxies™ composite will exhibit $T_g \leq 300^\circ\text{C}$ OR Flexural Modulus ≤ 15 GPa, indicating that bio-based chemical routes or HDCF carbonization yield a material structurally inferior to petroleum-based aerospace composites.

2 Safety Precautions

This protocol involves several highly hazardous materials and processes. All experiments must be conducted in a dedicated fume hood or inert atmosphere glove box with appropriate engineering controls. Standard laboratory PPE (nitrile gloves, safety goggles, flame-resistant lab coat) is mandatory.

Specific Hazard Areas

- **High-Pressure Synthesis:** Experiment C requires high-pressure hydrogenation (15 bar) using H₂ gas and catalyst. Ensure the reactor is pressure-rated (certified to ≥ 30 bar), subject to annual hydrostatic testing, and located in a blast-shielded environment with hydrogen sensors.
- **Corrosive Reagents and Intermediates:** Experiment A involves Gaseous HCl and strong NaOH solution, as well as highly toxic Epichlorohydrin (Bio-ECH) intermediate. This process is a known pathway for converting bio-derived glycerol.
 - Install a caustic scrubber system for HCl off-gas (minimum 2 M NaOH, 500 mL capacity).
 - Use a closed reactor with a pressure relief valve and rupture disk.
 - Install continuous ECH vapor monitoring (TWA limit: 0.5 ppm, STEL: 5 ppm).
 - An emergency shower and eyewash must be within 10 seconds of travel time.
 - Use an air-fed respirator if handling Bio-ECH outside of a high-flow fume hood (>100 fpm face velocity).
- **Thermal and Pyrolysis Hazards:** Experiments D, E, and F involve temperatures up to 1200 °C.
 - Implement strict burn prevention protocols.
 - Install emergency quench procedures (inert gas flooding).
 - Bio-Syngas is highly flammable (H₂ + CO content: 30-40%).
 - Use an explosion-proof burner system with flame arrestors.
 - Install combustible gas detectors (LEL alarms at 20% LEL).
- **Pyrophoric Catalysts:** Raney Nickel (Experiment C) must be stored and handled submerged under ethanol. Quench spent catalyst with water before disposal in a designated metal waste container.

3 Methods and Materials

The experimental plan is divided into two phases: Phase I (Biorefinery) focuses on the synthesis and purification of the 7 components, and Phase II (Formulation) covers the mixing and curing protocols for the final composite.

3.1 Hemp Biorefinery Component Sourcing and Map

3.2 Phase I: Synthesis of *Hempoxies*TM Components

Experiment A: Bio-Epichlorohydrin (Bio-ECH) Synthesis

Step 1: Hydrochlorination

1. Charge 500 mL glass pressure reactor with 100.0 g (1.086 mol) Hemp Glycerol.
2. Heat to 110 °C with stirring (300 rpm).
3. Bubble gaseous HCl at 50 mL/min for 4 hours (≈ 100 g, 2.74 mol; ratio 1:2.5). This step follows established chemistry for converting glycerol into dichloropropanol intermediates.
4. Cool to room temperature.
5. Expected: 1,3-Dichloropropan-2-ol (yield $\approx 85\%$).

Table 1: Mapping of *C. sativa* Fractions to Hemposies™ Components (14 → 7 Map)

ID	Fraction Name	Component	Function
1	Hemp Seed Oil	EHSO	Main Resin Matrix
2	Hemp Glycerol	Bio-ECH*	Epoxide Precursor
3	Hemp Carboxylic Acids	EHSO Synthesis	Epoxidation Catalyst
4	Hemp Phytic Acid	Q-MH Lignin Additive	Fire Retardant
5	Hemp Cystine	Q-MH Lignin Additive	Vitrimer Function
6	Hemp Pentosans	Hemp FGE & Amine	Diluent & Hardener
7	Organosolv Lignin	Q-MH Lignin	Hardener/Crosslinker
8	Residual Biomass	Biochar	Micro-Filler
9	Hemp Potash (K_2CO_3)	HDCNS Activator	Synthesis Agent
10	Long Bast Fiber	HDCF	Structural Fiber
11	Short Bast Fiber	HDCNS	Nano-Reinforcement
12	Pyrolysis Bio-Char	Biochar	Micro-Filler
13	Bio-Syngas	Process Energy	Heat Source
14	Bio-Oil (Aqueous)	Process Solvent	Byproduct/Recycle

synthesized from Fraction 2

Experiment B: Q-MH Lignin Synthesis

Step 1: Phosphorylation

1. Add 10.0 g Hemp Phytic Acid + 2.0 g triphenylphosphine to a solution of 100.0 g Organosolv Hemp Lignin in 400 mL DMF. Phytic acid is an effective bio-based flame retardant that acts by promoting char formation.
2. Heat to 140 °C for 3 hours under N_2 .
3. Monitor FTIR: P=O at 1250 cm^{-1} to 1280 cm^{-1} .

Step 3: Vitrimerization

1. Cool to 80 °C, filter NaCl.
2. Add 5.0 g Hemp Cystine + 8.0 g DCC. The disulfide bonds in cystine are introduced to enable thermally reversible cross-linking for self-healing properties.
3. React 4 hours at 80 °C.
4. Monitor FTIR: S-S at 500 cm^{-1} to 540 cm^{-1} .

Experiment D: Hemp Carbon Nanosheets

Step 2: Chemical Activation

1. Mix 75.0 g hydrochar + 300.0 g K_2CO_3 (1:4 ratio). This high-ratio impregnation with potassium carbonate is a proven method for creating high-surface-area carbons from biomass.
2. Transfer to an alumina crucible in a tube furnace under N_2 (500 mL/min).
3. Heat at 10 °C/min to 800 °C, hold 2 hours.
4. Cool under N_2 .
5. Wash with 1 M HCl, then water until pH 6-7.
6. Dry at 105 °C for 12 hours.
7. Expected HDCNS: 35 g to 40 g.

Experiment E: Hemp Carbon Fiber

Step 2: Stabilization

1. Place fiber tow in an air furnace.
2. Heat at $<1^\circ\text{C}/\text{min}$ to 250°C . This slow heating in air is a critical step to thermally stabilize the cellulose structure and prevent fiber fusion before carbonization.
3. Hold 2 hours, cool slowly.
4. Expected mass retention: 90%.

Step 3: Carbonization

1. Transfer to a tube furnace under N_2 (1 L/min).
2. Heat at $10^\circ\text{C}/\text{min}$ to 1200°C , hold 1 hour. This temperature is within the typical range for producing carbon fibers from hemp, which balances graphitization and mechanical properties.
3. Use Bio-Syngas combustion for heating.
4. Cool under N_2 .
5. Expected HDCF: 60 g to 70 g.

3.3 Phase II: Formulation and Curing

Experimental Groups (n=8 per group)

Table 2: Experimental Groups

Group	Formulation
Group 1	Full Hemposies™: EHSO/Q-MH/HDCF/HDCNS/Biochar
Group 2	HDCF/EHSO (commercial amine hardener)
Group 3	HDCF/EHSO/Q-MH (no HDCNS)
Group 4	EHSO/Q-MH Matrix Only (no fiber)
Control A	T-800H Carbon Fiber / Hexcel 8552 Epoxy
Control B	2 wt% GNP / DGEBA Epoxy

Cure Cycle

1. Gelation: Heat to 80°C at $2^\circ\text{C}/\text{min}$, hold 2 hours.
2. Intermediate: Ramp to 180°C at $2^\circ\text{C}/\text{min}$, hold 4 hours.
3. Post-Cure: Ramp to 280°C at $1^\circ\text{C}/\text{min}$, hold 4 hours (critical for $T_g > 300^\circ\text{C}$).
4. Cooling: Cool to 100°C at $<3^\circ\text{C}/\text{min}$, release vacuum, demold.
5. Total: ≈ 14 hours.

4 Data Collection and Analysis

Composite Characterization

Vitrimer Testing:

1. Test 4 specimens to failure (ASTM D790), record mean strength σ_0 .
2. Fracture remaining 4 specimens (load to 90% σ_0).
3. Sand fracture surfaces (600-grit), clean with acetone.

4. Weld: Align in a jig, apply 100 kPa pressure, heat to 180 °C for 30 min.
5. Cool slowly, re-test, calculate WE (%) = $(\sigma_{healed}/\sigma_0) \times 100$.
6. Target: WE > 70%. This target is ambitious but achievable for disulfide-based vitrimers.
7. Stress relaxation: DMA at 160 °C to 220 °C, calculate τ^* and E_a .

Statistical Analysis

- **Hypothesis:** $T_g > 300$ °C AND Flexural Modulus > 15 GPa.
- **Null:** $T_g \leq 300$ °C OR Modulus ≤ 15 GPa.
- **Significance:** $\alpha = 0.05$.

5 Future Work: Viability Assessment

Techno-Economic Analysis (TEA)

Following NREL guidelines, model cost per kg using Aspen Plus V11 for mass/energy balance. Benchmark vs. PAN-CF (50 \$/kg to 150 \$/kg).

Life Cycle Assessment (LCA)

Following ISO 14040/14044, cradle-to-gate using SimaPro 9.0 with Ecoinvent 3.8. Functional unit: 1 kg composite. Quantify Bio-Syngas offset ($\geq 80\%$ external natural gas). Target: < 50% GWP vs. conventional (30 kg to 40 kg CO₂-eq/kg). The GWP of conventional PAN-based carbon fiber composites is often cited in the range of 20-40 kg CO₂-eq per kg of composite, providing a clear benchmark.

6 Conclusion

This work aims to demonstrate a total closed-loop biorefinery system that valorizes 14 distinct fractions from *Cannabis sativa* into a 7-component, high-performance carbon fiber composite with integrated vitrimer self-healing functionality. The critical innovation—replacing moisture-sensitive cellulosic fibers with hemp-derived carbon fiber (HDCF) via 1200 °C carbonization—is intended to enable thermal stability and mechanical properties meeting aerospace-grade specifications. The path to commercialization requires scaling of continuous HDCF production, cost reduction in lignin synthesis, and comprehensive LCA/TEA validation.

7 References

References

- [1] Dumanlı, A. G., & Windle, A. H. (2012). Carbon fibres from cellulosic precursors: a review. *Journal of Materials Science*, 47(3), 1143-1155.
- [2] Li, Y., et al. (2020). Preparation of carbon fibers from hemp fibers via facile method. *Journal of Applied Polymer Science*, 137(19), 48791.
- [3] Fache, M., et al. (2016). Lignin-based epoxy resins: towards a sustainable alternative for petroleum-based polymers. *Green Chemistry*, 18(1), 57-70.
- [4] Thakur, V. K., et al. (2014). Progress in green polymer composites from lignin for structural applications: a review. *ACS Sustainable Chemistry & Engineering*, 2(5), 1072-1092.
- [5] Denissen, W., et al. (2016). Vitrimers: permanent organic networks with glass-like fluidity. *Chemical Reviews*, 116(3), 1182-1234.
- [6] Ma, S., et al. (2018). A robust, self-healing, and recyclable supramolecular polyurethane elastomer enabled by a dynamic disulfide bond. *Journal of Materials Chemistry A*, 6(19), 9814-9821.

- [7] Wang, H., et al. (2013). Interconnected carbon nanosheets derived from hemp for ultrafast supercapacitors with high energy. *ACS Nano*, 7(6), 5131-5141.
- [8] Karinen, R., & Krause, A. O. I. (2011). New bio-based chemicals—Glycerol conversion to epichlorohydrin. In *Bio-based chemicals* (pp. 165-188). Springer, Berlin, Heidelberg.
- [9] Costes, L., et al. (2017). Bio-based flame retardants: a review. *Materials Science and Engineering: R: Reports*, 117, 1-28.
- [10] Suzuki, T., & Takahashi, J. (2013). Life cycle assessment of carbon fiber reinforced plastics. In *Life Cycle Assessment of advanced production systems* (pp. 1-13).